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Reversal Q-Learning

Flow-Based Offline RL Trains Expressive Policies Without Backpropagating Through the Denoising Chain

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Flow-matching policies can now be trained from offline data without differentiating through their iterative denoising chains. *Reversal Q-learning* (RQL) treats each denoising step as an MDP action, reverses completed flows to generate synthetic on-policy trajectories, and applies standard Q-learning—beating prior flow-based offline RL methods on every D4RL locomotion task while eliminating the instability of backpropagation through time.

AT A GLANCE

Problem: Training an expressive flow-matching policy with offline RL requires either weak guidance distillation or expensive, unstable backpropagation through the full denoising chain.

Why it matters: Multimodal action distributions are common in robotics and navigation datasets; flow policies can represent them but existing RL training methods are ill-suited to offline settings.

Method: Treat denoising steps as MDP actions, reverse completed flows to obtain on-policy trajectories from offline data, and train with standard off-policy Q-learning plus a bias-variance reduction step.

Result: State-of-the-art on 12 D4RL locomotion tasks with faster and stabler training than BPTT-based methods.

Evidence: Systematic evaluation on D4RL medium, medium-replay, and medium-expert splits vs. IQL, TD3+BC, DQL, IDQL, SfBC.

The Big Picture

Offline reinforcement learning trains a policy from a fixed dataset of past experience without additional environment interaction. The challenge is simultaneously estimating a value function over out-of-distribution actions (hard) and representing multi-modal action distributions that arise when the dataset mixes several different behavioral strategies (also hard).

Flow matching—a continuous normalising flow that transforms noise into an action via an ordinary differential equation (ODE)—is a natural policy class for the multi-modal problem. Its generation process is smooth, invertible, and arbitrarily expressive. But plugging flow policies into offline RL has been awkward: distillation-based methods (SfBC, IDQL) under-use the value function, and direct RL fine-tuning via policy gradient requires differentiating through all T ODE steps, which is computationally expensive and numerically fragile.

RQL resolves this by lifting the problem into an *expanded* Markov decision process where each denoising step is its own atomic action. Standard off-policy Q-learning then suffices—no differentiation through the ODE solver, full use of the Q-function at every step.

Why Existing Approaches Fall Short

Guidance distillation (e.g., IDQL, SfBC) trains a diffusion model via behaviour cloning and then biases sampling towards high-Q regions at test time. The policy is never directly trained to maximise Q; it remains anchored to the behavioural prior.

BPTT fine-tuning (e.g., DQL) back-propagates a policy gradient through the full T-step ODE unrolling. This suffers from (i) exploding/vanishing gradients over long chains, (ii) prohibitive memory cost, and (iii) biased gradients because the ODE solver itself is not differentiable in the strict sense.

Neither exploits the structure of the flow ODE for off-policy learning.

Core Mechanism

Expanded MDP. Discretise $[0,1]$ into T steps with $\Delta t = 1/T$. Define an augmented state $\tilde{s}_k = (s, x_{k\Delta t})$ where s is the environment state and x_k is the current latent. The action at step k is $\delta_k = v_\theta(\tilde{s}_k) \cdot \Delta t$. Reward is $R_T = Q^*(s, x_1)$ at the final step and zero elsewhere.

Reversal. Given an offline pair $(s, a^*) \in \mathcal{D}$, reconstruct the full latent trajectory by running the ODE *backwards*: $x^*_{k-1} = x^*_k - v_\theta(x^*_k, k\Delta t; s) \cdot \Delta t$, with $x^*_T = a^*$. This converts each offline sample into a complete expanded-MDP episode.

Q-learning. Apply Bellman updates over these synthetic trajectories. The flow policy is updated by maximising $Q_\phi(\tilde{s}_k, v_\theta(\tilde{s}_k))$ at each step—a single-step gradient, never multi-step.

Technical Formulation

A flow matching policy defines an ODE:

$$\frac{dx_t}{dt} = v_\theta(x_t, t; s), \quad x_0 \sim \mathcal{N}(0, I), \quad a = x_1.$$

The expanded-MDP Q-function satisfies:

$$Q^*(\tilde{s}_k, \delta_k) = \begin{cases} Q^*(s, x_{k+1}) & k < T-1 \\ r + \gamma\, \mathbb{E}_{s'}[V^*(s')] & k = T-1 \end{cases}$$

The Q-network is trained with:

$$\mathcal{L}_Q(\phi) = \mathbb{E}\Big[\big(Q_\phi(\tilde{s}_k, \delta_k^*) - y_k\big)^2\Big]$$

and the policy is updated by:

$$\mathcal{L}_\pi(\theta) = -\mathbb{E}_{k,s,x_0}[Q_\phi(\tilde{s}_k,\, v_\theta(\tilde{s}_k))]\,.$$

Crucially, gradients of $\mathcal{L}_\pi$ pass only through the *single-step* output $v_\theta(\tilde{s}_k)$, not through the ODE solver.

Learning or Inference Procedure

1. Sample $(s, a^*) \in \mathcal{D}$.
2. Run the reverse ODE to obtain $\{x_k^*\}_{k=0}^T$.
3. Compute Bellman targets $\{y_k\}$ using current Q_ϕ.
4. Update Q_ϕ via $\mathcal{L}_Q$.
5. Update v_θ via $\mathcal{L}_\pi$ (single-step gradient only).
6. At test time: run the forward ODE from $x_0 \sim \mathcal{N}(0, I)$ for T steps to generate action $a = x_1$.

What the Guarantee Says

The paper proves that, under the expanded MDP, the optimal Q-function Q^* decomposes across denoising steps exactly as stated, and that the reversal procedure yields unbiased estimates of the on-policy return. A bias-variance reduction technique—weighting intermediate step returns by the learned Q-function rather than bootstrapping all the way from the final step—reduces variance by a factor proportional to T without introducing bias.

Experimental Findings

Evaluated on the 12 D4RL locomotion tasks (HalfCheetah, Hopper, Walker2d at medium / medium-replay / medium-expert quality), RQL achieves:

- Highest average normalised score vs. IQL, TD3+BC, DQL, IDQL, SfBC.
- Particularly large gains on medium-expert splits, where multi-modal data benefits most from an expressive policy.
- Stable training curves without the loss spikes observed in DQL (BPTT).
- Wall-clock training time comparable to IQL despite operating over an expanded T-step MDP.

Ablations and Interpretation

Number of steps T: Performance plateaus around $T = 10$; $T = 1$ reduces to standard Q-learning with a one-step denoiser, which underperforms.

Reversal vs. forward noise: Replacing the reversal with forward noising (add Gaussian noise to a^*) degrades performance by ≈ 8 normalised score on average, confirming that faithfully reconstructing the latent trajectory is important.

Q-weighting at intermediate steps: Removing the bias-variance reduction and using terminal Q-values only increases variance and slows convergence by $\approx 2\times$.

Reference

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